

Program : Diploma in Automation and Robotics	
Course Code : 6331A	Course Title: Advanced Robotics
Semester: 6	Credits : 4
Course Category: Program core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To introduce the concepts of mobile robotic system, its components.
- Familiarize with the robot trajectory planning and control

Course Prerequisites:

Foundations of Robotics, Mechanics of Robotics

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Familiarize with different types of mobile robots, kinematic models, motion control and sensors for mobile robots.	15	Understanding
CO2	Familiarize with robot learning process.	14	Understanding
CO3	Familiarize with mobile robot path planning.	15	Understanding
CO4	Learn robot programming and motion control of robots.	14	Understanding
	Series Test	2	

CO – PO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1							
CO2	3	2					
CO3	3	2					
CO4		2					

3-Strongly mapped 2-Moderately mapped 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Familiarize with different types of mobile robots, kinematic models, motion control and sensors for mobile robots.		
M1.01	Introduction to mobile robots	5	Understanding
M1.02	Advances in mobile robots.	5	Understanding
M1.03	Discuss mobile robot kinematics.	5	Understanding
Contents: Introduction of Mobile Robotics, Mechanics and Locomotion: A brief history of mobile robotics, applications and market. Recent advances in the mobile robotics for RISE (Risky Intervention and Surveillance Environment) applications, Locomotion, Key issues in locomotion, legged, wheeled and aerial mobile robots. Mobile Robot Kinematics: Introduction, kinematic models and constraints, mobile robot workspace, beyond basic kinematics, motion control (kinematic control).			
CO2	Familiarize with robot learning process.		
M2.01	Discuss about robot learning process.	5	Understanding
M2.02	Explain the components of robot learning process	5	Understanding
M2.03	Discuss robot learning case studies.	4	
	Series Test – I	1	
Contents: Perception, robotics Architectures and Robot Learning: Sensors Classification, sensor characterization, wheel/motor encoders, heading/orientation sensors, ground based beacons, active ranging, motion/speed sensors, vision based sensors. Low level control, Control architectures, software frameworks, Robot Learning, case studies of learning robots.			
CO3	Familiarize with mobile robot path planning.		
M3.01	Explain mobile robot localization	8	Understanding
M3.02	Discuss different path planning algorithms.	7	Understanding
Contents: Mobile Robot Localization: Introduction, the challenge of localization: Noise and aliasing, to localize or not to localize: localization-based navigation versus programmed solutions, map representation, probabilistic map, map-based localization, autonomous map building. Planning and navigation: Planning and reaction, obstacle avoidance, D* algorithm, A* algorithm, Navigation architecture, case studies.			

CO4	Learn robot programming and motion control of robots.		
M4.01	Familiarize robot programming	7	
M4.02	Discuss the motion control of mobile robots.	7	
	Series Test – II	1	
Contents: Robot Programming: Methods of robot programming- Textual and Leadthrough, WAIT, SIGNAL and DELAY commands, Capabilities and limitations of leadthrough programming, Robot language structure, Motion, sensor and end effectors commands, Programming examples. Motion Control: The control problem, Joint space control, Decentralized control, Computed torque feed forward control, Centralized control, PD Control with gravity compensation, Inverse dynamics control, Operational space control.			

Text / Reference

T/R	Book Title/Author
T1	Roland Siegwart&Illah R. Nourbakhsh, “Introduction to autonomous mobile robots”, Prentice Hall of India, 2004.
T2	Howie Choset, Kevin M. Lynch, Seth Hutchinson, George A. Kantor, Wolfram Burgard, Lydia E. Kavrakiand Sebastian Thrun, "Principles of Robot motion: Theory, Algorithm and Implementations", MIT Press.
R1	George A. Bekey “Autonomous Robots” MIT Press.
R2	Robert. J. Schilling , “Fundamentals of robotics – Analysis and control”, Prentice Hall of India 1996.
R3	R K Mittal and I J Nagrath, “Robotics and Control”, Tata McGraw Hill, New Delhi,2003.